Lecture L01
Probability and Bayesian Statistics

From Bayesian statistics to probabilistic machine learning

Probabilistic Machine Learning 2027-1

1970-01-01

YOUR NOTES

Section 1

A familiar modeling language

YOUR NOTES

Lecture L01 — Probability & Statistics

The Beta-Binomial model

Observed

the coin tosses $y_1, ..., y_N$

Unknown

the coin bias θ

$\theta \sim \text{Beta}(\alpha, \beta), \quad \alpha, \beta > 0$

$y_i \mid \theta \sim \text{Bernoulli}(\theta), \quad i = 1, ..., N$

Conditionally independent tosses given θ

Heads: $y_i = 1$; tails: $y_i = 0$.

$N_1 = \sum_{i=1}^{N} y_i$

$N_0 = N - N_1$

- A statistical model has data and unknown parameters
- How could observed data (y) have been generated.
- What does it tell us about the parameters (θ)
- N_1 and N_0 count heads and tails in the observed sample

STUDY NOTE

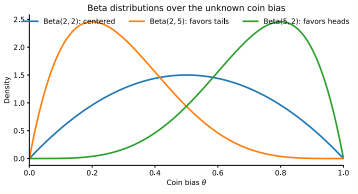
The coin gives us a complete miniature probabilistic model: a prior for the unknown bias and a sampling law for observed tosses. Separating what is unknown from what is observed establishes the language we will use to ask what data tell us.

YOUR NOTES

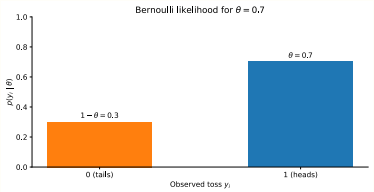
Lecture L01 — Probability & Statistics

Models are built out of distributions

$$\theta \sim \text{Beta}(\alpha, \beta)$$



$$y_i \mid \theta \sim \text{Bernoulli}(\theta)$$
$$i = 1, \ldots, N$$



- Beta for continuous variables on $[0, 1]$.
- Given θ: independent Bernoulli tosses; Binomial for the count N_1.
- Distributions are parameterized families

1. A familiar modeling language

S1.03 · Slide 4/55

STUDY NOTE

The Beta and Bernoulli distributions specify different objects: uncertainty about the shared coin bias and toss variation conditional on that bias. These are parameterized probability families rather than free-standing curves.

YOUR NOTES

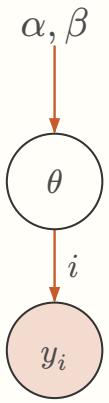
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Models can be drawn as graphs

Carried forward S3 · Coin model

$\theta \sim \text{Beta}(\alpha, \beta)$

$y_i \mid \theta \sim \text{Bernoulli}(\theta)$



Nodes represent quantities; shading marks observed data.
Arrows specify the modeled conditional relationships.

1. A familiar modeling language

S1.04 · Slide 5/55

STUDY NOTE

The graph makes the dependence assumption explicit: all tosses share one bias, and become independent once that bias is fixed. This structure lets the observations jointly inform the same unknown rather than treating each toss as a separate problem.

YOUR NOTES

Lecture L01 — Probability & Statistics

Conjugacy permits simple calculations

Carried forward S3 · Coin model and counts

$\theta \sim \text{Beta}(\alpha, \beta)$

$$N_1 = \sum_{i=1}^{N} y_i, \quad N_0 = N - N_1$$

Bayes' rule

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)} \propto p(D \mid \theta)p(\theta)$$

$$p(D \mid \theta) = \prod_{i=1}^{N} p(y_i \mid \theta) \propto \theta^{N_1}(1-\theta)^{N_0}$$

Prior $\theta \sim \text{Beta}(\alpha, \beta)$

Likelihood $p(D \mid \theta) \propto \theta^{N_1}(1-\theta)^{N_0}$

Posterior $\theta \mid D \sim \text{Beta}(\alpha + N_1, \beta + N_0)$

> Conjugacy keeps the posterior in the prior's distribution family.

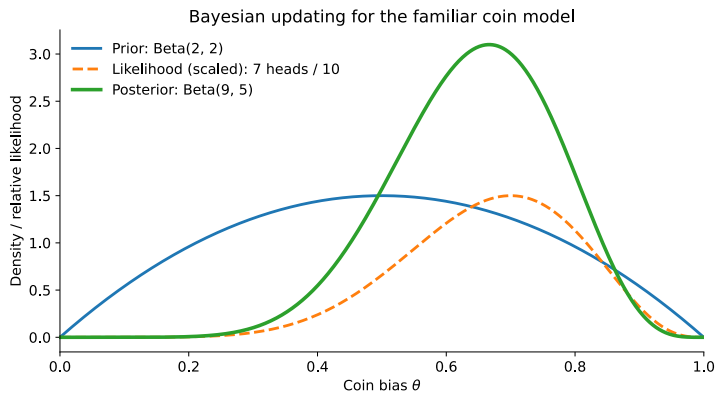
STUDY NOTE

Bayes' rule updates the bias distribution using the observed head and tail counts; Beta conjugacy keeps the answer in the same family. This first exact calculation shows how a probabilistic model turns data into a distributional answer.

YOUR NOTES

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The posterior is a distribution representing uncertainty



Bayesian updating for the familiar coin model

- The prior and the posterior are both probability distributions
- Posterior width represents remaining uncertainty
- In this example, the posterior is narrower than the prior

1. A familiar modeling language

S1.06 · Slide 7/55

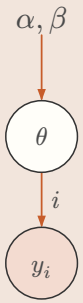
STUDY NOTE

The result of conditioning is a distribution over plausible biases, not just one estimate. For these data the posterior curve narrows, making uncertainty itself part of the answer?a distinction we will keep when later choosing point predictions.

YOUR NOTES

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Section 1 summary: A familiar modeling language



$$\theta \sim \text{Beta}(\alpha, \beta), \quad y_i \mid \theta \sim \text{Bernoulli}(\theta)$$
$$\theta \mid D \sim \text{Beta}(\alpha + N_1, \beta + N_0)$$

TAKE HOME

- A model specifies uncertain quantities and their relationships.
- Bayes combines prior and likelihood into a posterior distribution for the unknown parameter.

STUDY NOTE

We began with observed tosses and an uncertain bias, encoded their relation in a graph, and used Bayes' rule to update the bias distribution. This small example established the pattern of specifying a model, identifying an observation, and answering a conditional query.

YOUR NOTES

Section 2

From statistics to machine-learning problems

Same probabilistic language, different organizing questions

YOUR NOTES

Lecture L01 — Probability & Statistics

Statistics focuses on the unknown parameters

Carried forward S6 · Bayes and conjugacy

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}$$

> The familiar coin query concerns an unknown parameter.
> A prediction problem also specifies the output required for a new input.

STUDY NOTE

The coin calculation asked what data imply about an unknown parameter. A machine-learning problem may instead ask what output to produce for a new case, so the question must now specify the available experience, the required prediction, and how it will be judged.

YOUR NOTES

Lecture L01 — Probability & Statistics

Machine learning grew out of computer science



- A computational task is an **input/output contract**
  - Produce a function f that maps input x to output y
- Implementation is secondary to the requirement
- Multiple implementations may be possible
- Machine learning developed in this context

STUDY NOTE

Computer science often starts by specifying which outputs a computation must produce for which inputs. Machine learning asks how experience can construct an appropriate function rather than requiring us to program that mapping directly.

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Early machine learning grew out of classification problems

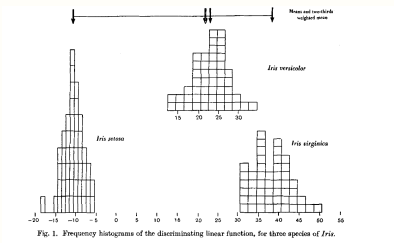


Fig. 1. Frequency histograms of the discriminating linear function, for three species of *Iris*.

Fisher (1936), Fig. 1, *The Use of Multiple Measurements in Taxonomic Problems*

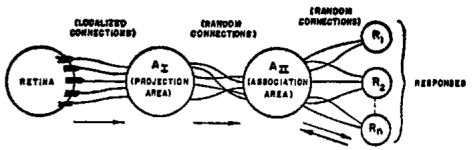


FIG. 1. Organization of a perceptron.

Rosenblatt (1958), Fig. 1, *The Perceptron*

- Fisher's discriminant and Rosenblatt's perceptron share an interface
  - complex input → class or response
- Machine learning asks how data can produce the required function f

2. From statistics to machine learning

S2.04 · Slide 12/55

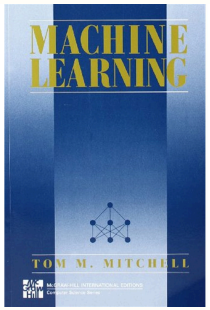
STUDY NOTE

Classification uses past examples to learn a response that can be applied to a new input. These early examples shift the lecture from inference about a model's unknowns to the practical requirement of making useful predictions on unseen cases.

YOUR NOTES

Lecture L01 — Probability & Statistics

Machine learning defined



Tom M. Mitchell, *Machine Learning* (1997), cover

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.

Training	Task	Evaluation
$f_T : \mathrm{E} \rightarrow f_A$	$f_A : \mathrm{In} \rightarrow \mathrm{Out}$	$f_P : (\mathrm{In}, \mathrm{Out}) \rightarrow \mathrm{P}$
E: Experience	f_A: Task function	P: Performance

- Training f_T uses experience E to construct the task function f_A
  - Its aim is to improve performance according to P

2. From statistics to machine learning

S2.05 · Slide 13/55

STUDY NOTE

Mitchell explicitly separates the task T, experience E, and performance P. Training uses the experience to improve measured task performance. We will make these abstract ingredients concrete in the Iris classification and loss examples.

YOUR NOTES

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Classification: Iris flowers



Setosa



Versicolor



Virginica

Murphy (2025), Book 1, Fig. 1.1

Table 1.1: subset of the Iris design matrix

index	sl	sw	pl	pw	label
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
dots	dots	dots	dots	dots	dots
50	7.0	3.2	4.7	1.4	Versicolor
dots	dots	dots	dots	dots	dots
149	5.9	3.0	5.1	1.8	Virginica

sl: sepal length; sw: sepal width; pl: petal length; pw: petal width

Murphy (2025), Book 1, Table 1.1

> The 150 cases are labeled; each has four measurements $x_n \in \mathbb{R}^4$ and species label y_n. Training on $D = \{(x_n, y_n)\}$ gives a predictor f_D; for a new input x_* it returns $\hat{y} = f_D(x_*)$.

2. From statistics to machine learning

S2.06 · Slide 14/55

STUDY NOTE

The Iris example makes the prediction objects concrete: one flower supplies a four-measurement input, and its species is the label available during training but unknown for a new flower. It grounds the abstract supervised-learning notation in identifiable cases.

YOUR NOTES

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A loss function scores a prediction

Carried forward Iris classification and Mitchell task definition

$\hat{y} = f_D(x_*)$

Predicted label $\hat{y}$; observed true label y.

$$\ell_{01}(y, \hat{y}) \triangleq \mathbb{I}\{y \neq \hat{y}\}$$

$$\mathbb{I}\{y \neq \hat{y}\} = \begin{cases} 0 \text{ correct label} \\ 1 \text{ misclassified} \end{cases}$$

> The criterion scores prediction errors after the true label is known.
> We want useful performance on future cases.

STUDY NOTE

A prediction task needs a criterion as well as inputs and outputs. Zero-one loss scores whether the chosen species is right once the true label is known, making errors on future cases the performance concern.

YOUR NOTES

Lecture L01 — Probability & Statistics

A class model gives probabilities and a label decision

Supplied or previously fitted parameters θ.

Class prior: $p_\theta(c)$

Class-conditional feature law: $p_\theta(x \mid c)$

Classify a case

$$p_\theta(c \mid x) \propto p_\theta(x \mid c) p_\theta(c)$$

$$\hat{c} \triangleq \arg\max_c p_\theta(c \mid x)$$

Carried forward S15 · Zero-one loss

$$\ell_{01}(c, \hat{c}) = \mathbb{I}\{c \neq \hat{c}\}$$

Class probabilities and a selected class are different outputs; under zero-one loss, choose a most-probable class.

STUDY NOTE

A probabilistic classifier gives a distribution over possible labels; under zero-one loss, choosing its most probable class converts that distribution into a decision. This connects the Bayesian query to a prediction while keeping uncertainty, decision, and scoring distinct.

YOUR NOTES

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Section 2 summary: From statistics to machine learning

Training	Task	Evaluation
$f_T : \mathrm{E} \to f_A$ E: experience	$f_A : \mathrm{In} \to \mathrm{Out}$ f_A: learned function	$f_P : (\mathrm{In}, \mathrm{Out}) \to P$ P: performance

TAKE HOME

- Machine learning asks how data can produce the required function f
- A learning problem is defined by three ingredients: task, experience, and performance

2. From statistics to machine learning

S2.09 · Slide 17/55

STUDY NOTE

The section connects a computer-science task specification to statistical prediction: experience supports learning, the task specifies required behavior, and performance supplies the criterion of success. Classification and zero-one loss made the three ingredients concrete.

YOUR NOTES

Section 3

Classes of machine learning problems

Experience regimes, task classes, and performance criteria

YOUR NOTES

Lecture L01 — Probability & Statistics

Experience describes what the learner observes

Supervised

Paired inputs and targets: $D = \{(x_n, y_n)\}$.

Unsupervised

Inputs without supplied target labels: $D = \{x_n\}$.

Self-supervised

A supplied transformation forms input/target pairs: $x_n \to f(x_n)$, or $f(x_n) \to x_n$.

Reinforcement / interaction

Observe state x_t, choose action a_t, then observe reward r_{t+1} and state x_{t+1}.

> Experience describes what the learner gets to observe.

3. ML taxonomies

S3.02 · Slide 19/55

STUDY NOTE

The Iris pairs are one kind of learning experience: inputs arrive with supplied labels. Other settings provide unlabelled inputs, targets constructed from inputs, or feedback from interaction. Naming the available signal is the first step in describing the learning setup.

YOUR NOTES

Lecture L01 — Probability & Statistics

Task describes what the system must do

Prediction

$x_* \to \hat{y} = f_D(x_*)$

Map a new input to an output.

Generation

$D = \{x_n\} \to p_D(x)$

$x_* \sim p_D(x)$

Image-only experience supplies examples for learning the generative law.

Discovery

Find useful structure or a representation in the data.

Action

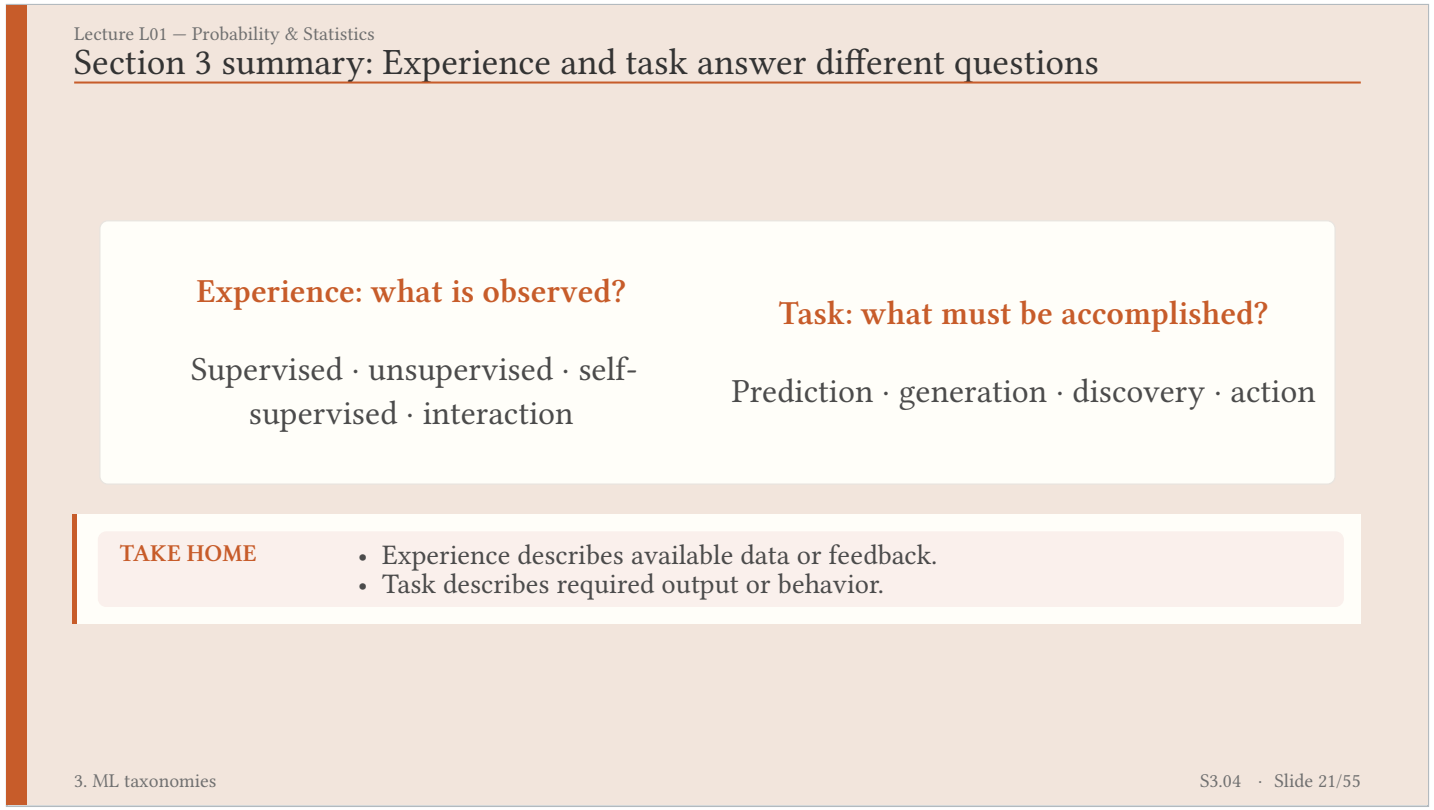
Choose actions to achieve useful future outcomes, judged through reward.

A task specifies the required output or behavior and what counts as useful performance.

STUDY NOTE

Experience alone does not determine what the system must do. Prediction, generation, discovery, and action request different outputs and success criteria, so the task must be stated separately from the data or feedback available.

YOUR NOTES



STUDY NOTE

A learning problem has two distinct parts: what the learner observes and what behavior it must produce. The same observations can serve different tasks, and similar tasks can use different feedback; keeping both parts explicit prevents a collection of data from standing in for a complete problem.

YOUR NOTES

Section 4

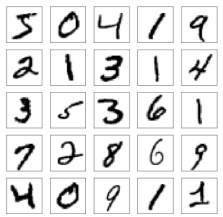
From datasets to learning problems

The same dataset can pose different learning problems

YOUR NOTES

Lecture L01 — Probability & Statistics

MNIST: the same taxonomy, one dataset



$x_n \in \{0, \ldots, 255\}^{28 \times 28}$
$y_n \in \{0, \ldots, 9\}$

Murphy (2025), Book 1, Fig. 1.12a

Prediction
$$\hat{y} = f_D(x_*)$$

Discovery
$$\hat{z} = f_D(x_*)$$

Generation
$$x_* \sim p_D(x \mid c)$$

- **Prediction:** use labeled digit images to predict the class of a new image
- **Discovery:** seek a representation $\hat{z}$ of an image rather than its digit label
- **Generation:** draw a new image from $p_D(x \mid c)$, possibly for a requested class c

4. Problem examples

S4.02 · Slide 23/55

STUDY NOTE

MNIST is a collection of digit images and labels, not a single prescribed learning problem. We might classify a new image, seek structure among images, or generate images; this concrete example tests why the task must be chosen in addition to the available experience.

YOUR NOTES

Lecture L01 — Probability & Statistics

A specified model turns an image question into a query

One supplied or previously fitted joint $p_\theta(x, c)$; parameters θ are fixed.

Classify an image

Observe: new image x_*. **Unknown:** class c.

$p_\theta(c \mid x_*)$

$\hat{c} \triangleq \arg\max_c p_\theta(c \mid x_*)$

Carried forward S15 · Zero-one loss

$\ell_{01}(c, \hat{c}) = \mathbb{I}\{c \neq \hat{c}\}$

Generate an image for a class

Specify: requested class c. **Produce:** new image x. $x \sim p_\theta(x \mid c)$

Under zero-one loss, choose a class with largest posterior probability.
The model and what is observed or requested determine the query; learning θ is separate.

4. Problem examples

S4.03 · Slide 24/55

STUDY NOTE

To make the image questions computable, we must supply a model. A joint distribution for image and class supports classification by conditioning on the image and class-conditional generation by conditioning on the class; the different outputs are queries to the same specified model.

YOUR NOTES

Lecture L01 — Probability & Statistics

Section 4 summary: From datasets to learning problems

Observed data



$D = \{(x_n, c_n)\}_{n=1}^{N}$

Specify a model: $p_\theta(x, c)$ with supplied or fitted θ.

Then specify the question:

Classification : $p_\theta(c \mid x_*)$

Conditional generation : $x_* \sim p_\theta(x \mid c)$

TAKE HOME

- Observations alone do not determine which task the learner must solve.
- Given a model, the task identifies the probabilistic query; estimating model parameters is a separate operation.

STUDY NOTE

MNIST supplies observed inputs and class labels, but possible tasks ask for different outputs. The model p_theta (x,c) supports classification and class-conditional generation through different conditional distributions. Model fitting is conceptually separate from these queries.

YOUR NOTES

Section 5

Joint Gaussian distributions and geometry

Describe and transform joint uncertainty with a mean and covariance

YOUR NOTES

Lecture L01 — Probability & Statistics

Gaussians: a concrete example

Univariate $\mathcal{N}(x|\mu,\sigma^2) = \frac{\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}} \quad \sigma^2 > 0$

Multivariate $\mathcal{N}(\boldsymbol{x}|\mu,\Sigma) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^T\Sigma^{-1}(\boldsymbol{x}-\mu)\right)}{(2\pi)^{\frac{D}{2}}\,|\Sigma|^{\frac{1}{2}}} \quad \Sigma \succ 0$

Recall Linear algebra: quadratic forms
$\Sigma \succ 0$ means **positive definite**: $\Sigma = \Sigma^T$ and $\boldsymbol{v}^T\Sigma\boldsymbol{v} > 0$ for every nonzero $\boldsymbol{v} \in \mathbb{R}^D$.

Murphy, Book 2 §2.3.1.1, Eq. 2.71; scalar case: $D = 1$

- Univariate: mean μ and positive variance $\sigma^2 > 0$
- Multivariate: mean $\mu \in \mathbb{R}^D$ and positive-definite covariance $\Sigma \in \mathbb{R}^{D\times D}$
- Gaussian means, marginals, and conditionals have closed forms

5. Gaussian geometry

S5.02 · Slide 27/55

STUDY NOTE

The Gaussian is a tractable model for dependent continuous quantities: its mean gives location, and its covariance gives spread and dependence. This concrete family lets us answer the earlier questions about probabilistic queries with exact calculations.

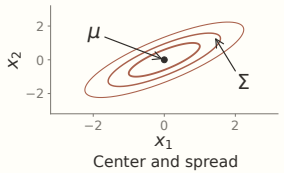
YOUR NOTES

Lecture L01 — Probability & Statistics

The three operations in multivariate Gaussian systems

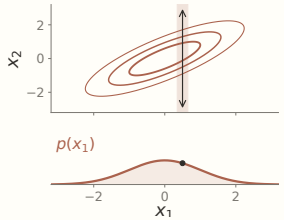
One joint Gaussian: $p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma)$

Summarize



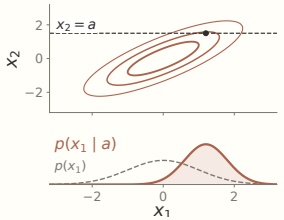
$$\mu = \int \boldsymbol{x} p(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x}$$

Marginalize



$$p(x_1) = \int p(x_1, x_2) \, \mathrm{d}x_2$$

Condition



$$p(x_1 \mid x_2 = a) = \frac{p(x_1, a)}{\int p(u, a) \, \mathrm{d}u}$$

Course figures generated by `gaussian_geometry_figures.ipynb`

- **Summarize:** describe the center and spread with μ and Σ
- **Marginalize:** integrate over the coordinate we ignore
- **Condition:** fix an observed coordinate and normalize the slice

STUDY NOTE

Before calculating with a joint Gaussian, we must distinguish the answer requested: a summary, a marginal that ignores an unobserved coordinate, or a conditional that uses an observed value. This distinction determines which exact Gaussian rule the later calculations need to establish.

YOUR NOTES

Lecture L01 — Probability & Statistics

Covariance summarizes joint variation

For $\boldsymbol{X} = (X_1, \ldots, X_D)^T$ with mean $\mu = E[\boldsymbol{X}]$

$$\Sigma_{ij} \triangleq E\big[(X_i - \mu_i)(X_j - \mu_j)\big]$$

$$\Sigma \triangleq E\big[(\boldsymbol{X} - \mu)(\boldsymbol{X} - \mu)^T\big] = E\big[\boldsymbol{X}\boldsymbol{X}^T\big] - \mu\mu^T$$

$$\Sigma_{ii} = \mathrm{Var}(X_i) = \sigma_i^2$$

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}$$

Murphy, Book 1 §3.1.1–3.1.2

> Diagonal entries are variances; off-diagonal entries describe joint variation. Zero covariance does not imply independence in general.

5. Gaussian geometry

S5.04 · Slide 29/55

STUDY NOTE

Covariance records how Gaussian coordinates vary together, while correlation expresses that association without units. The cross-covariance terms matter because they will determine whether observing one coordinate changes what we know about another.

YOUR NOTES

Lecture L01 — Probability & Statistics

Affine maps carry means and covariances

Given $E[X] = \mu$, $\mathrm{Cov}(X) = \Sigma$ and fixed A, b.

Affine map and its mean

$$Y = AX + b$$
$$E[Y] = A\mu + b$$

Center the variables

$$D = X - \mu, \quad E[D] = 0$$
$$Y - E[Y] = AD$$

Transform the covariance

$$\mathrm{Cov}(Y) = E\big[(AD)(AD)^T\big]$$
$$= AE\big[DD^T\big]A^T = A\Sigma A^T$$

Use $(AD)^T = D^T A^T$; fixed matrices commute with expectation.

Gaussian affine closure

$$X \sim \mathcal{N}(\mu, \Sigma)$$
$$Y = AX + b$$
$$Y \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$$

Fixed T; set $A = T$, $b = 0$:
$$TX \sim \mathcal{N}(T\mu, T\Sigma T^T)$$

Murphy, Book 1 §3.1.1, Eq. 3.5; Gaussian affine transformations

5. Gaussian geometry

S5.05 · Slide 30/55

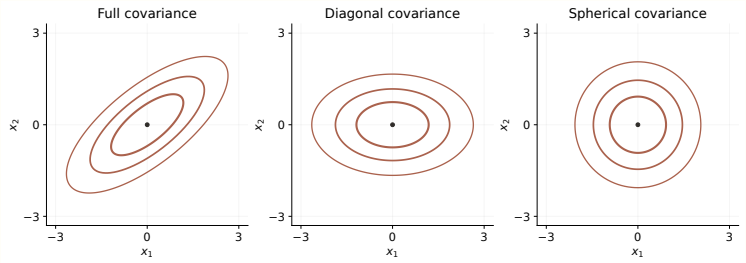
STUDY NOTE

An affine map shifts the mean and, after centering, transforms covariance by taking fixed matrices outside expectation. These moment identities need only finite second moments. Gaussian affine closure additionally identifies the full transformed law; its matrix form will justify Gaussian marginals and later state-measurement models.

YOUR NOTES

Lecture L01 — Probability & Statistics

Covariance controls the contour shape



Course rendering of Murphy, Book 2, Fig. 2.8

- A full covariance can rotate the principal axes
- Diagonal covariance removes linear correlation in these coordinates
- Spherical covariance assigns the same variance in every direction

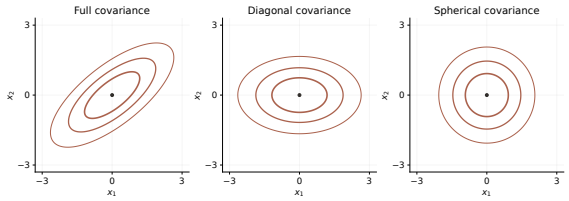
STUDY NOTE

Gaussian contours make covariance visible: their orientation shows dependence, while their axis lengths show spread. This geometric picture gives meaning to the covariance blocks that will govern how much an observation can change an estimate.

YOUR NOTES

Lecture L01 — Probability & Statistics

Section 5 summary: Joint Gaussian geometry



$$\boldsymbol{X} \sim \mathcal{N}(\mu, \Sigma)$$
$$A\boldsymbol{X} + b \sim \mathcal{N}\left(A\mu + b, A\Sigma A^T\right)$$

TAKE HOME

- Mean and covariance locate and shape dependent uncertainty.
- Affine maps carry the moments and preserve the Gaussian family.

STUDY NOTE

The Gaussian's mean and covariance described location and dependent spread, and affine maps transformed those parameters while preserving Gaussianity. We also distinguished summarizing, marginalizing, and conditioning as different questions about a joint distribution.

YOUR NOTES

Section 6

Marginals and conditionals of a Gaussian

Ignore coordinates or use observations to update uncertainty

YOUR NOTES

Lecture L01 — Probability & Statistics

Partition the variables and their parameters together

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma), \quad X_1 \in \mathbb{R}^p, \quad X_2 \in \mathbb{R}^q$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \succ 0$$

$$\Sigma_{12} \in \mathbb{R}^{p \times q}, \quad \Sigma_{21} = \Sigma_{12}^T$$

Murphy, Book 2 §2.3.1.3, Eq. 2.75

- Model parameters are supplied; variable blocks may be observed or missing.
- Diagonal blocks describe within-group variation; cross blocks describe variation between groups.

STUDY NOTE

To answer a query about one part of a joint Gaussian, we partition the variables, means, and covariance in matching blocks. This makes the observed-versus-unobserved choice precise and identifies the cross-covariance that can carry information between the blocks.

YOUR NOTES

Lecture L01 — Probability & Statistics

Marginalization keeps the matching blocks

Carried forward S35 · Joint Gaussian blocks
$\boldsymbol{x} \sim \mathcal{N}(\mu, \Sigma)$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$
$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$
$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

Marginal proof

Select x_1: $P = [I \;\; 0]$, $\quad x_1 = P\boldsymbol{x}$

Carried forward S31 · Affine Gaussian rule
$A\boldsymbol{x} + b \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$
$$P\mu = \mu_1$$
$$P\Sigma P^T = [I \;\; 0] \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I \\ 0 \end{bmatrix}$$
$$= \Sigma_{11}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing x_2 integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of x_1

STUDY NOTE

If the second block is unobserved, selecting the first is an affine map, so its marginal is Gaussian with the matching mean and covariance. This is the first exact query rule: ignoring a coordinate does not use any particular value of it.

YOUR NOTES

Lecture L01 — Probability & Statistics

Conditioning shifts the mean by an observed deviation

Carried forward S35 · Supplied joint Gaussian blocks

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma), \quad \Sigma \succ 0$$

Define residual

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$
$$R = X_1 - \mu_1 - B(X_2 - \mu_2)$$

Residual and independence

$E[R] = 0$
$\mathrm{Cov}(R, X_2) = \Sigma_{12} - B\Sigma_{22} = 0$
(R, X_2) is jointly Gaussian by affine closure
Thus $R \perp X_2$

Conditional mean at fixed x_2

$$X_1 = \mu_1 + B(X_2 - \mu_2) + R$$
$$E[R \mid X_2 = x_2] = E[R] = 0$$
$$\mu_{1|2} = E[X_1 \mid X_2 = x_2]$$
$$= \mu_1 + B(x_2 - \mu_2)$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- The conditional mean shifts with the observed deviation $x_2 - \mu_2$.
- Gaussianity turns zero cross-covariance into the independence needed here.

STUDY NOTE

Conditioning differs because an observed value can shift our estimate of the first block. Subtracting the part predicted from the second block leaves a residual uncorrelated?and, in this joint Gaussian, independent?of the observation; that independence justifies the mean update.

YOUR NOTES

Lecture L01 — Probability & Statistics

Conditioning removes explained covariance

Carried forward Previous slide · Independent residual

$B = \Sigma_{12}\Sigma_{22}^{-1}$

$X_1 = \mu_1 + B(X_2 - \mu_2) + R, \quad R \perp X_2$

Add the two independent covariance contributions

$$\Sigma_{11} = B\Sigma_{22}B^T + \mathrm{Cov}(R)$$

$$S \triangleq \mathrm{Cov}(R) = \Sigma_{11} - B\Sigma_{22}B^T$$

Substitute B

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.79

STUDY NOTE

Because the residual R is independent of the observed block X2, their covariance contributions add. Subtract the part predictable from X2 to identify S as the residual covariance. The removed term is positive semidefinite, so conditioning never increases covariance in any direction.

YOUR NOTES

Lecture L01 — Probability & Statistics

The independent residual yields the full conditional distribution

Carried forward Gaussian model and independent residual

$X \sim \mathcal{N}(\mu, \Sigma), \quad \Sigma \succ 0$
$R = X_1 - \mu_1 - B(X_2 - \mu_2), \quad R \perp X_2$
$E[R] = 0, \quad \mathrm{Cov}(R) = S$

The residual retains its Gaussian distribution

$R \sim \mathcal{N}(0, S)$ by affine Gaussian closure

$(R \mid X_2 = x_2) \sim \mathcal{N}(0, S)$ by independence

Translate by the conditional mean

$(X_1 \mid X_2 = x_2) \sim \mathcal{N}(\mu_1 + B(x_2 - \mu_2), S)$

Murphy, Book 2 §2.3.1.3, Eqs. 2.78–2.79

- The conditional mean depends on the value x_2; covariance S does not.

6. Marginals and conditionals

S6.06 · Slide 38/55

STUDY NOTE

The independent residual is Gaussian by affine closure, and its distribution remains unchanged when X2 is observed. Reconstructing X1 adds the observation-dependent mean to that residual. This proves the full conditional Gaussian law, not merely the conditional mean and covariance formulas.

YOUR NOTES

Lecture L01 — Probability & Statistics

The bivariate formula gives intuition

$$x_1, x_2 \in \mathbb{R}$$

$$(X_1 \mid X_2 = x_2) \sim \mathcal{N}\left(\mu_{1|2}, \sigma^2_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$$

$$\sigma_{1|2} = \sigma_1 \sqrt{1 - \rho^2}$$

Murphy, Book 1 §3.2.4; Book 2 §2.3.1.3

- Positive ρ moves the estimate in the observed direction
- $\rho = 0$ leaves both mean and variance unchanged
- Larger $|\rho|$ leaves less conditional variance

STUDY NOTE

For two scalar variables, correlation makes the general conditional rule easier to read: its sign controls the mean shift, and its magnitude controls the variance reduction. This specializes the block result into an interpretable rule before applying it to numbers.

YOUR NOTES

Lecture L01 — Probability & Statistics

One observed coordinate can change both the estimate and its uncertainty

Supplied joint model

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}\right)$$

Recorded: $X_2 = 1.5$; **missing:** X_1

$X_2 = 1.5$ **recorded**

$$(X_1 \mid X_2 = 1.5) \sim \mathcal{N}(0.8(1.5), 1 - 0.8^2) = \mathcal{N}(1.2, 0.36)$$

X_2 **unobserved**

$$X_1 \sim \mathcal{N}(0, 1)$$

Course-constructed example; Gaussian conditioning: Murphy, Book 2 §2.3.1.3, Eq. 2.78

- With X_2 unobserved, marginalize it to obtain $p(X_1)$.
- Both calculations use supplied μ, Σ; neither fits them.
- Conditioning changes the mean $0 \to 1.2$ and variance $1 \to 0.36$.

STUDY NOTE

With one coordinate observed, conditioning updates both the estimate and uncertainty of the missing one; without that observation, marginalization gives its unchanged distribution. The numerical example checks that we choose the query from the available information before doing arithmetic.

YOUR NOTES

Lecture L01 — Probability & Statistics

The Schur complement is the residual covariance

Carried forward S37–S39 · Conditional Gaussian

$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$
$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$
$(X_1 \mid X_2 = x_2) \sim \mathcal{N}\left(\mu_{1|2}, \Sigma_{1|2}\right)$

Carried forward S37–S39 · Independent residual

$R = X_1 - \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}(X_2 - \mu_2)$
$R \sim \mathcal{N}\left(0, \Sigma_{1|2}\right)$
The residual R is independent of X_2

$$S = \Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Remove the part predicted by X_2:

$$\mathrm{Cov}\left[\begin{pmatrix} R \\ X_2 \end{pmatrix}\right] = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.91, 2.102–2.103

- S measures the uncertainty left after subtracting the conditional mean
- Block elimination leaves S: the **Schur complement of Σ_{22} in Σ**

6. Marginals and conditionals

S6.09 · Slide 41/55

STUDY NOTE

The Schur complement is the covariance left in the residual after the observed block's contribution has been removed. Naming it connects the earlier conditional derivation to the standard block-Gaussian form; the block-diagonal covariance records why residual and observation separate.

YOUR NOTES

Lecture L01 — Probability & Statistics

The joint quadratic separates into conditional and marginal parts

Carried forward S38 · Residual covariance
$r = x_1 - \mu_{1|2}, \quad S = \Sigma_{1|2}$

$$p(x_1, x_2) = p(x_1 \mid x_2)p(x_2)$$

Stated quadratic identity

$$(\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu) = r^T S^{-1} r + (x_2 - \mu_2)^T \Sigma_{22}^{-1} (x_2 - \mu_2)$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.94–2.103

- The first quadratic describes x_1 after seeing x_2
- The second describes the marginal distribution of x_2

STUDY NOTE

The joint quadratic splits into a residual part and a part for the observed block. This algebra mirrors the factorization into conditional and marginal laws, giving another view of the same Gaussian decomposition without repeating the conditional proof.

YOUR NOTES

Lecture L01 — Probability & Statistics

Section 6 summary: Marginals and conditionals of a Gaussian

Marginalize

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$
$$x_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$$

Condition

$$p(x_1|x_2) = \mathcal{N}\left(x_1|\mu_{1|2}, \Sigma_{1|2}\right)$$
$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

TAKE HOME

- Marginalizing x_2 integrates over all its possible values
- Conditioning shifts the mean using the observed deviation
- Conditioning cannot increase variance in any direction

STUDY NOTE

When the second block was unobserved, marginalization retained the first block's mean and covariance; when it was observed, conditioning shifted the mean and removed explained covariance. The independent Gaussian residual established why both answers are complete probability laws.

YOUR NOTES

Section 7

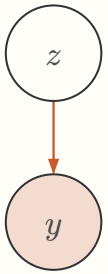
Linear Gaussian systems

Inference from noisy linear measurements

YOUR NOTES

Lecture L01 — Probability & Statistics

A linear Gaussian system is a model of state and measurement



Latent state

Observed

A prior and a measurement law

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$
$$p(z, y) = p(z)p(y \mid z)$$

$$y = Wz + b + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Omega)$$

Noise ε is independent of the state z.

Murphy, Book 2 §2.3.2, Eqs. 2.118–2.119

- This model specifies how an unknown state generates a noisy measurement
- The prior and noise covariances satisfy $\Sigma_z \succ 0, \Omega \succ 0$
- Here the parameters $\mu_z, \Sigma_z, W, b, \Omega$ are given

STUDY NOTE

A linear Gaussian system uses a prior over a hidden state and a noisy linear measurement rule to create a joint model of state and observation. It turns Gaussian conditioning into a recognizable inference question: what can a sensor reading tell us about an unseen state?

YOUR NOTES

Lecture L01 — Probability & Statistics

One linear Gaussian system supports different questions

Carried forward Linear Gaussian model

$z \sim \mathcal{N}(\mu_z, \Sigma_z)$
$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$

Condition

$p(z \mid y) = \frac{p(y \mid z)p(z)}{p(y)}$

Summarize

$E[g(z) \mid y] = \int g(z)p(z \mid y)\,\mathrm{d}z$

Marginalize

$p(y) = \int p(y \mid z)p(z)\,\mathrm{d}z$

Murphy, Book 2 §2.3.2

- **Condition:** infer z after observing y
  - Bayesian posterior
- **Summarize:** use $E[z \mid y]$ as an estimate of z
  - Bayesian estimator
- **Marginalize:** predict y before observing it by integrating out z
  - Prior predictive

STUDY NOTE

Once a linear Gaussian measurement model is specified, we can ask for a posterior distribution, summarize that posterior by an expectation, or marginalize over the state to predict readings. These are different operations on one model, not different model families.

YOUR NOTES

Lecture L01 — Probability & Statistics

A linear Gaussian model defines a joint Gaussian

Carried forward S46 · Linear Gaussian model

$z \sim \mathcal{N}(\mu_z, \Sigma_z)$
$y = Wz + b + \varepsilon$
$\varepsilon \sim \mathcal{N}(0, \Omega), \quad \varepsilon \perp z$

Carried forward S31 · Affine Gaussian rules

$u = Ax + c$
$E[u] = AE[x] + c$
$\mathrm{Cov}[u] = A\ \mathrm{Cov}[x]A^T$

Compute the joint moments

$\mu_y = E[Wz + b + \varepsilon] = W\mu_z + b$

Independent noise gives zero cross terms:

$\Sigma_{zy} = \mathrm{Cov}[z, Wz + \varepsilon] = \Sigma_z W^T$

$\Sigma_{yy} = \mathrm{Cov}[Wz + \varepsilon] = W\Sigma_z W^T + \Omega$

Joint Gaussian

Independence makes (z, ε) jointly Gaussian:

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_z & \Sigma_{zy} \\ \Sigma_{zy}^T & \Sigma_{yy} \end{pmatrix} \right)$$

$\Sigma_{zz} = \Sigma_z, \quad \Sigma_{yz} = \Sigma_{zy}^T$

Murphy, Book 2 §2.3.2.1, Eq. 2.120

- Start from the model and affine-Gaussian rules
- Independent noise gives the joint moments; joint Gaussianity enables both queries.

STUDY NOTE

The model equations yield the joint Gaussian blocks required by the conditioning rule. State uncertainty and sensor noise both enter measurement covariance, while cross-covariance records what the measurement shares with the state; that shared variation makes inference from a reading possible.

YOUR NOTES

Lecture L01 — Probability & Statistics

Bayes' rule for Gaussians

Carried forward Computed state-measurement joint

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right)$$

$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$

$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$

$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Carried forward Established Gaussian conditioning law

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Apply: $x_1 = z$, $x_2 = y$

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}(y - \mu_y)$$

$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Murphy, Book 2 §2.3.2.2, Eq. 2.121

- Apply the Gaussian conditioning rule to the joint blocks
- The mean responds to observed minus predicted measurement
- The covariance subtracts the uncertainty explained by the measurement

STUDY NOTE

Apply the conditional Gaussian law to the joint covariance blocks of the state and measurement. The resulting posterior mean and covariance come directly from substituting the two blocks, not from a new theorem.

YOUR NOTES

Lecture L01 — Probability & Statistics

Bayes' rule as an update formula

Carried forward Previous slide · Gaussian posterior

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}(y - \mu_y)$$
$$\mu_y = W\mu_z + b$$
$$\Sigma_{zy} = \Sigma_z W^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Define the gain

$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$
$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$

Murphy, Book 2 §2.3.2.2, Eqs. 2.122–2.124

- K is the gain: it converts a measurement discrepancy into a state correction
- Posterior estimate = prior estimate + gain × prediction error

STUDY NOTE

The gain K packages the cross-covariance and measurement precision. With K defined, the posterior mean is prior mean plus gain times prediction error, a useful computational interpretation of the Gaussian conditioning formula.

YOUR NOTES

Lecture L01 — Probability & Statistics

The scalar gain measures relative uncertainty

Carried forward S49 · Gaussian update

$\mu_{z|y} = \mu_z + K(y - \mu_y)$

$K = \Sigma_{zy}\Sigma_{yy}^{-1}$

Scalar specialization

$y, z \in \mathbb{R}, \quad W = 1, \quad b = 0$

$\Sigma_z = \sigma_z^2 > 0, \quad \Omega = \sigma_\varepsilon^2 > 0$

$z \sim \mathcal{N}(\mu_z, \sigma_z^2)$

$y = z + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$

ε is independent of z

Predicted mean

$\mu_y = E[z + \varepsilon] = \mu_z$

Cross-covariance

$\Sigma_{zy} = \mathrm{Cov}[z, z + \varepsilon] = \sigma_z^2$

Measurement variance

$\Sigma_{yy} = \mathrm{Var}[z + \varepsilon] = \sigma_z^2 + \sigma_\varepsilon^2$

$$K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2} \quad \rightarrow \quad \mu_{z|y} = \mu_z + K(y - \mu_z)$$

Murphy, Book 2 §2.3.2.2; scalar specialization of Eqs. 2.121–2.123

- K is the fraction of $y - \mu_z$ used to correct μ_z.

STUDY NOTE

In one dimension, the gain is set by relative state and sensor uncertainty. The scalar formula clarifies the general update: a measurement has more influence when it is precise compared with what was already known about the state.

YOUR NOTES

Lecture L01 — Probability & Statistics

Measurement noise controls the scalar update

Context S51 · Scalar model, gain and correction

$$y = z + \varepsilon,\ \varepsilon \perp z$$
$$K = \frac{\sigma_z^2}{\sigma_z^2+\sigma_\varepsilon^2}$$
$$\mu_{z|y} = \mu_z + K(y-\mu_z)$$

Hold μ_z, σ_z^2, and observed y fixed; vary σ_ε^2.

Measurement regime	Gain	Posterior mean
Precise: $\sigma_\varepsilon^2 \ll \sigma_z^2$	$K \approx 1$	$\mu_{z\|y} \approx y$
Equal: $\sigma_\varepsilon^2 = \sigma_z^2$	$K = \frac{1}{2}$	$\mu_{z\|y} = \frac{\mu_z+y}{2}$
Noisy: $\sigma_\varepsilon^2 \gg \sigma_z^2$	$K \approx 0$	$\mu_{z\|y} \approx \mu_z$

- More measurement noise lowers K, so the posterior mean moves less toward y
- With measurement variance fixed, larger prior variance raises K

STUDY NOTE

The three noise regimes test that interpretation: a nearly noiseless reading dominates, equal uncertainties balance prior and observation, and a noisy reading changes little. Holding the prior and measurement fixed isolates what sensor quality contributes to the update.

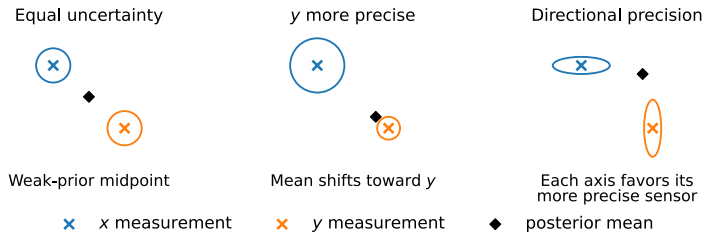
YOUR NOTES

Lecture L01 — Probability & Statistics

Sensor fusion combines measurements according to their uncertainty

Two-sensor model

$$Z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$X \mid Z = z \sim \mathcal{N}(z, \Sigma_x), \quad Y \mid Z = z \sim \mathcal{N}(z, \Sigma_y), \quad X \perp Y \mid Z$$



Weak-prior-limit qualification of Murphy, Book 2 §2.3.2.3, Fig. 2.10; source: `sensor_fusion_figure.ipynb`

- Equal isotropic noise: weak-prior midpoint; finite prior adds mean and precision.
- Higher precision moves the mean toward its measurement; anisotropy is directional; correlation changes covariance.

7. Linear Gaussian systems

S7.09 · Slide 52/55

STUDY NOTE

Two conditionally independent sensor readings inform the same hidden state, and their contributions depend on precision, including direction-specific precision. The weak-prior midpoint is only a special case; sensor fusion extends the one-reading gain idea to several sources of evidence.

YOUR NOTES

Lecture L01 — Probability & Statistics

The same model can also predict measurements

Carried forward S48 · Constructed joint Gaussian

$p(z, y) = p(z)p(y \mid z)$

$Y \mid Z = z \sim \mathcal{N}(Wz + b, \Omega)$

Carried forward S36 · Marginalization rule

$p(y) = \int p(z, y)\, \mathrm{d}z$

$= \int p(y \mid z)p(z)\, \mathrm{d}z$

$$p(y) = \mathcal{N}\left(y \mid \mu_y, \Sigma_{yy}\right)$$
$$\mu_y = W\mu_z + b, \quad \Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Murphy, Book 2 §2.3.2.2, Eq. 2.129

- Before observing y, marginalization gives the distribution of possible measurements
- These are the same μ_y and Σ_{yy} that appear inside the conditioning update

STUDY NOTE

Before seeing a reading, prediction averages over possible hidden states rather than plugging in one estimate. This produces measurement uncertainty from both the state and sensor noise, giving the forward counterpart to the state update.

YOUR NOTES

Lecture L01 — Probability & Statistics

Section 7 summary: Linear Gaussian systems

Model and joint blocks

$$Z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$Y \mid Z = z \sim \mathcal{N}(Wz + b, \Omega)$$
$$\mu_y = W\mu_z + b, \quad C = \Sigma_z W^T$$
$$V_y = W\Sigma_z W^T + \Omega$$

Inverse and forward queries

Inverse: $K = CV_y^{-1}$

$$p(z|y) = \mathcal{N}\left(z \mid \mu_z + K(y - \mu_y), \Sigma_z - KC^T\right)$$

Forward: $p(y) = \mathcal{N}(y \mid \mu_y, V_y)$

TAKE HOME

- Conditioning gives an uncertainty-weighted state update; marginalization gives possible measurements before observation.
- The same joint model supplies both queries using its cross-covariance and measurement covariance.

STUDY NOTE

The noisy sensor example completed the lecture's path from a learning question to a joint probabilistic model and an exact query. Once the state-measurement covariance blocks are known, observing a reading calls for conditioning; asking what readings are possible calls for marginalization.

YOUR NOTES

Lecture L01 — Probability & Statistics

Lecture 1: four visual anchors

One dataset, several questions

A dataset is not a task.



$x \to \hat{y}$ Prediction
$x \to \hat{z}$ Discovery
$x_* \sim p_\theta(x \mid c)$ Generation

Three probability operations

The task determines the operation.

Summarize $E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$
Marginalize $p(x) = \int p(x, y)\,\mathrm{d}y$
Condition $p(x \mid y) = \frac{p(x,y)}{p(y)}$

Gaussian conditioning

Observation changes estimate and uncertainty.

$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$
$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$

Sensor fusion

Uncertainty controls how evidence is combined.



Visual anchors from Lecture 1: MNIST tasks, three probability operations, Gaussian conditioning, and sensor fusion

STUDY NOTE

This is a lecture-wide closing synthesis after the Linear Gaussian systems section summary. The four anchors link learning questions, probabilistic operations, conditional uncertainty, and evidence-weighting sensor fusion.

YOUR NOTES